INCH-POUND
MIL-PRF-55365/11
28 February 2003

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, POLARIZED ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY, STYLES CWR19 AND CWR29

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-55365.

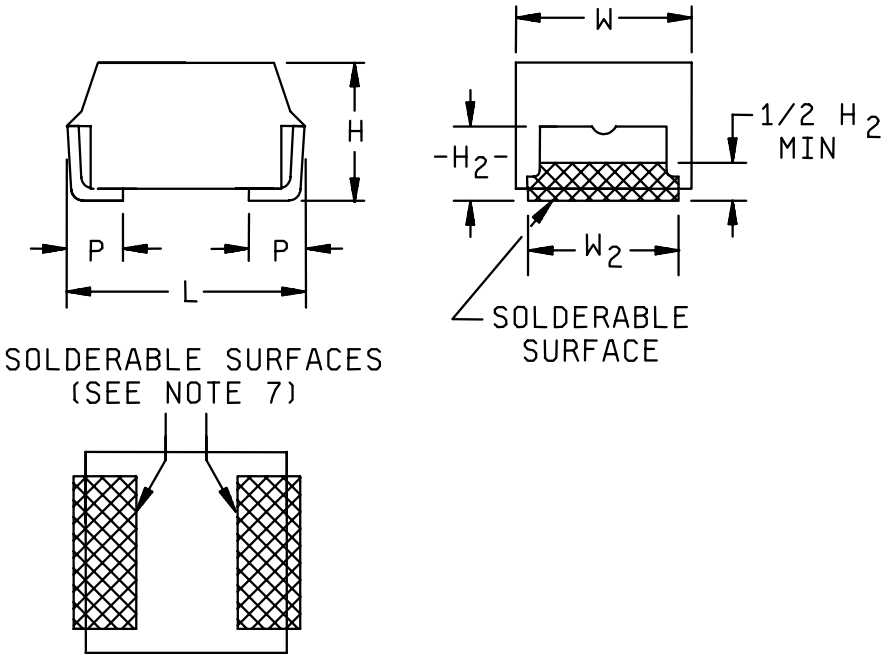


FIGURE 1. Style CWR19 and CWR29 (molded) capacitors.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Case size	L ± .015 (0.38)	W ±.015 (0.38)	H ±.015 (0.38)	P +.010 (0.25) -.005 (0.13)	W_2	H_2 min
A	.100 (2.54)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±0.13)	.030 (0.76)
B	.150 (3.81)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±0.13)	.030 (0.76)
C	.200 (5.08)	.050 (1.27)	.050 (1.27)	.030 (0.76)	.050 ±.005 (1.27±0.13)	.030 (0.76)
D	.150 (3.81)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005,-.010 (2.41+0.13, -0.25)	.030 (0.76)
E	.200 (5.08)	.100 (2.54)	.050 (1.27)	.030 (0.76)	.095+.005,-.010 (2.41+0.13, -0.25)	.030 (0.76)
F	.220 (5.59)	.135 (3.43)	.070 (1.78)	.030 (0.76)	.130±.005 (3.30±0.13)	.040 (1.02)
G	.265 (6.73)	.110 (2.79)	.110 (2.79)	.050 (1.27)	.105±.005 (2.67±0.13)	.060 (1.52)
H	.285 (7.24)	.150 (3.81)	.110 (2.79)	.050 (1.27)	.145+.005. -.020 (3.68+0.13, -0.51)	.060 (1.52)
X	.273 (6.93)	.213 (5.41)	.108 (2.74)	.047 (1.19)	.120 ± .005 (3.05 ± 0.13)	.048 (1.22)

NOTES:
1. Dimensions are in inches. Metric (millimeter) equivalents are given in parenthesis for general information only.
2. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
3. The anode (+) terminal shall be identified by a marking on the case.
4. The termination width (W_2) shall be maintained to the top of the solderable area.
5. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
6. When solder coated terminations are required, add an additional .015 inch (0.38 mm) to the above listed tolerances for “L”, “W”, “H”, “P”, “W_2”, and H_2 for each case size.

FIGURE 1. Style CWR19 and CWR29 (molded) capacitors - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Termination finish: In accordance with termination finishes B, C, H, or K of MIL-PRF-55365.

DC rated voltage: See table I or table II. Above +85°C, voltage derating is required (see MIL-PRF-55365).

Operating temperature range: -55°C to +125°C.

Product level designator: In accordance with MIL-PRF-55365.

DC leakage (DCL): See table I or table II.

Capacitance: See table I or table II.

Capacitance tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I or table II.

Equivalent series resistance (ESR) at 100 kHz: In accordance with MIL-PRF-55365 (see table I or table II).

Resistance to soldering heat: In accordance with MIL-PRF-55365.

Stability at low and high temperatures: In accordance with MIL-PRF-55365.

Step 1 (+25°C):	DCL	-	See table I or table II.
	Capacitance	-	Shall not exceed tolerance of table I or table II value.
	DF	-	See table I or table II.
Step 2 (-55°C):	ΔCapacitance	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I or table II.
Step 3 (+25°C):	DCL	-	See table I or table II.
	ΔCapacitance	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I or table II.
Step 4 (+85°C):	DCL	-	See table I or table II.
	ΔCapacitance	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I or table II.
Step 5 (+125°C):	DCL	-	See table I or table II.
	ΔCapacitance	-	Shall not exceed ±15 percent of initial +25°C measurement.
	DF	-	See table I or table II.
Step 6 (+25°C):	DCL	-	See table I or table II.
	ΔCapacitance	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I or table II.

Surge voltage: In accordance with MIL-PRF-55365.

DCL	-	See table I or table II.
ΔCapacitance	-	Within ±5 percent of initial value.
DF	-	See table I or table II.

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Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C:	DCL	- See table I or table II.
	ΔCapacitance	- Shall not exceed ±5 percent of initial measured value.
	DF	- See table I or table II.
At 85°C:	DCL	- See table I or table II.
At 125°C:	DCL	- See table I or table II.

10,000 hours:

At 25°C:	DCL	- 200 percent of initial requirement.
	ΔCapacitance	- Shall not exceed ±5 percent of initial measured value.
	DF	- See table I or table II.
At 85°C:	DCL	- 200 percent of initial requirement.
At 125°C:	DCL	- 200 percent of initial requirement.

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TABLE I. Style CWR19 characteristics.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR19C-335--A-	4	3.3	1	10	12	6	8	8	12	A
CWR19C-475--A-	4	4.7	1	10	12	6	8	8	12	A
CWR19C-685--A-	4	6.8	1	10	12	6	8	8	12	A
CWR19C-685--C-	4	6.8	1	10	12	6	8	8	5.5	C
CWR19C-106--B-	4	10	1	10	12	8	10	10	8	B
CWR19C-156--B-	4	15	1	10	12	8	10	10	8	B
CWR19C-226--B-	4	22	1	10	12	8	10	10	8	B
CWR19C-226--D-	4	22	1	10	12	8	10	12	4	D
CWR19C-336--D-	4	33	2	20	24	8	10	12	4	D
CWR19C-336--E-	4	33	2	20	24	8	10	12	3	E
CWR19C-476--E-	4	47	2	20	24	8	10	12	3	E
CWR19C-686--E-	4	68	3	30	36	8	10	12	3	E
CWR19C-107--F-	4	100	4	40	48	10	12	12	2	F
CWR19C-157--G-	4	150	6	60	72	10	12	12	1	G
CWR19C-227--G-	4	220	8	80	96	10	12	12	1	G
CWR19C-337--H-	4	330	10	100	120	10	12	12	0.9	H
CWR19D-335--A-	6	3.3	1	10	12	6	8	8	12	A
CWR19D-475--A-	6	4.7	1	10	12	6	8	8	12	A
CWR19D-475--C-	6	4.7	1	10	12	6	8	8	5.5	C
CWR19D-685--B-	6	6.8	1	10	12	6	8	8	8	B
CWR19D-106--B-	6	10	1	10	12	6	8	8	8	B
CWR19D-156--B-	6	15	1	10	12	8	10	10	8	B
CWR19D-156--D-	6	15	1	10	12	8	10	12	5	D
CWR19D-156--E-	6	15	1	10	12	8	10	12	3	E
CWR19D-226--D-	6	22	1	10	12	6	8	8	5	D
CWR19D-226--E-	6	22	2	20	24	8	10	12	3.5	E
CWR19D-336--E-	6	33	2	20	24	6	8	8	3.5	E
CWR19D-476--F-	6	47	3	30	36	8	10	12	3.5	F
CWR19D-686--F-	6	68	4	40	48	10	12	12	1.5	F
CWR19D-686--G-	6	68	4	40	48	10	12	12	1	G
CWR19D-107--G-	6	100	6	60	72	10	12	12	1.1	G
CWR19D-157--G-	6	150	10	100	120	10	12	12	1.1	G
CWR19D-227--H-	6	220	10	100	120	10	12	12	0.9	H
CWR19D-337--H-	6	330	20	200	240	10	12	12	0.9	H
CWR19F-225--A-	10	2.2	1	10	12	6	8	8	12	A
CWR19F-335--A-	10	3.3	1	10	12	6	8	8	12	A
CWR19F-335--C-	10	3.3	1	10	12	6	8	8	5.5	C
CWR19F-475--B-	10	4.7	1	10	12	6	8	8	8	B
CWR19F-475--C-	10	4.7	1	10	12	6	8	8	5.5	C
CWR19F-685--B-	10	6.8	1	10	12	6	8	8	8	B
CWR19F-685--C-	10	6.8	1	10	12	6	8	8	5.5	C
CWR19F-685--D-	10	6.8	1	10	12	6	8	8	5	D
CWR19F-106--B-	10	10	1	10	12	8	10	10	8	B
CWR19F-106--C-	10	10	1	10	12	6	8	8	5.5	C
CWR19F-106--D-	10	10	1	10	12	6	8	8	4	D
CWR19F-106--E-	10	10	1	10	12	6	8	8	3.5	E
CWR19F-156--D-	10	15	2	20	24	6	8	8	5	D
CWR19F-156--E-	10	15	2	20	24	8	10	10	3	E
CWR19F-226--E-	10	22	3	30	36	8	10	10	2	E
CWR19F-336--F-	10	33	3	30	36	8	10	10	1.5	F
CWR19F-476--F-	10	47	4	40	48	10	12	12	1.5	F
CWR19F-476--G-	10	47	4	40	48	10	12	12	1	G
CWR19F-686--G-	10	68	6	60	72	10	12	12	1.1	G
CWR19F-107--G-	10	100	10	100	120	10	12	12	1.1	G

See footnote at the end of table.

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TABLE I. Style CWR19 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR19F-107--H-	10	100	10	100	120	10	12	12	0.9	H
CWR19F-157--H-	10	150	15	150	180	10	12	12	0.9	H
CWR19F-157--X-	10	150	15	150	180	10	12	12	0.9	X
CWR19F-227--H-	10	220	20	200	240	10	12	12	0.9	H
CWR19H-105--A-	15	1	1	10	12	6	8	8	15	A
CWR19H-155--A-	15	1.5	1	10	12	6	8	8	15	A
CWR19H-225--A-	15	2.2	1	10	12	6	8	8	15	A
CWR19H-225--C-	15	2.2	1	10	12	6	8	8	5.5	C
CWR19H-335--B-	15	3.3	1	10	12	6	8	8	9	B
CWR19H-475--B-	15	4.7	1	10	12	6	8	8	5	B
CWR19H-475--C-	15	4.7	1	10	12	6	8	8	5.5	C
CWR19H-475--D-	15	4.7	1	10	12	6	8	8	6	D
CWR19H-685--D-	15	6.8	1	10	12	6	8	8	6	D
CWR19H-685--E-	15	6.8	1	10	12	8	10	12	3	E
CWR19H-106--D-	15	10	2	20	24	6	8	8	6	D
CWR19H-106--E-	15	10	2	20	24	6	8	8	4	E
CWR19H-156--E-	15	15	2	20	24	6	8	8	4	E
CWR19H-156--F-	15	15	2	20	24	8	10	10	3	F
CWR19H-226--F-	15	22	3	30	36	8	10	10	3	F
CWR19H-336--F-	15	33	5	50	60	6	8	8	3	F
CWR19H-336--G-	15	33	6	60	72	8	10	10	1.1	G
CWR19H-476--G	15	47	10	100	120	8	10	10	1.1	G
CWR19H-476--H-	15	47	10	100	120	8	10	10	0.9	H
CWR19H-686--G-	15	68	10	100	120	8	10	10	1.1	G
CWR19H-686--H-	15	68	10	100	120	8	10	10	0.9	H
CWR19H-107--H-	15	100	15	150	180	10	12	12	0.9	H
CWR19J-684--A-	20	0.68	1	10	12	6	8	8	15	A
CWR19J-105--A-	20	1	1	10	12	6	8	8	15	A
CWR19J-155--B-	20	1.5	1	10	12	6	8	8	9	B
CWR19J-155--C-	20	1.5	1	10	12	6	8	8	6	C
CWR19J-225--B-	20	2.2	1	10	12	6	8	8	9	B
CWR19J-335--D-	20	3.3	1	10	12	6	8	8	6	D
CWR19J-475--E-	20	4.7	1	10	12	6	8	8	6	E
CWR19J-685--E-	20	6.8	2	20	24	6	8	8	5	E
CWR19J-106--E-	20	10	2	20	24	6	8	8	5	E
CWR19J-106--F-	20	10	2	20	24	6	8	8	3	F
CWR19J-156--F-	20	15	3	30	36	6	8	8	3	F
CWR19J-226--G-	20	22	4	40	48	6	8	8	2.5	G
CWR19J-336--H-	20	33	6	60	72	8	10	10	0.9	H
CWR19J-476--H-	20	47	10	100	120	8	10	10	0.9	H
CWR19J-476--X-	20	47	10	100	120	8	10	10	0.9	X
CWR19K-474--A-	25	0.47	1	10	12	6	8	8	15	A
CWR19K-105--B-	25	1	1	10	12	6	8	8	10	B
CWR19K-105--C-	25	1	1	10	12	6	8	8	6.5	C
CWR19K-225--D-	25	2.2	1	10	12	6	8	8	6	D
CWR19K-335--E-	25	3.3	1	10	12	6	8	8	4	E
CWR19K-685--F-	25	6.8	2	20	24	6	8	8	3	F
CWR19K-156--G-	25	15	4	40	48	6	8	8	1.4	G
CWR19K-226--G-	25	22	6	60	72	6	8	8	1.4	G
CWR19K-226--H-	25	22	6	60	72	6	8	8	0.9	H
CWR19K-226--X-	25	22	6	60	72	6	8	8	0.9	X
CWR19K-336--H-	25	33	10	100	120	8	10	10	0.9	H
CWR19K-336--X-	25	33	10	100	120	8	10	10	0.9	X

See footnote at the end of table.

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TABLE I. Style CWR19 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR19M-334--A-	35	0.33	1	10	12	6	8	8	22	A
CWR19M-684--C-	35	0.68	1	10	12	6	8	8	10	C
CWR19M-685--G-	35	6.8	3	30	36	6	8	8	1.5	G
CWR19M-106--H-	35	10	4	40	48	8	10	10	0.9	H
CWR19M-156--X-	35	15	6	60	72	6	8	8	0.9	X
CWR19N-474--C-	50	0.47	1	10	12	6	8	8	8	C

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, product level designator, and surge current option letter. If surge current is not required, the last dash shall be replaced with the letter Z.

TABLE II. Style CWR29 characteristics.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR29C-225--A-	4	2.2	1	10	12	6	8	8	4	A
CWR29C-335--A-	4	3.3	1	10	12	6	8	8	6	A
CWR29C-475--A-	4	4.7	1	10	12	6	8	8	6	A
CWR29C-475--B-	4	4.7	1	10	12	6	8	8	3.2	B
CWR29C-685--A-	4	6.8	1	10	12	6	8	8	6	A
CWR29C-685--C-	4	6.8	1	10	12	6	8	8	2.2	C
CWR29C-106--B-	4	10	1	10	12	8	10	10	3.2	B
CWR29C-106--D-	4	10	1	10	12	8	8	10	1.3	D
CWR29C-156--B-	4	15	1	10	12	8	10	10	3.2	B
CWR29C-156--E-	4	15	1	10	12	8	10	12	1	E
CWR29C-226--B-	4	22	1	10	12	8	10	10	3.2	B
CWR29C-226--D-	4	22	1	10	12	8	10	12	1.3	D
CWR29C-336--D-	4	33	2	20	24	8	10	12	1.3	D
CWR29C-336--E-	4	33	2	20	24	8	10	12	0.9	E
CWR29C-336--F-	4	33	2	20	24	8	10	12	0.6	F
CWR29C-476--E-	4	47	2	20	24	8	10	12	0.9	E
CWR29C-686--E-	4	68	3	30	36	8	10	12	0.9	E
CWR29C-686--G-	4	68	3	30	36	10	12	12	0.275	G
CWR29C-107--F-	4	100	4	40	48	10	12	12	0.550	F
CWR29C-107--H-	4	100	4	40	48	10	12	12	0.180	H
CWR29C-157--G-	4	150	6	60	72	10	12	12	0.250	G
CWR29C-227--G-	4	220	8	80	96	10	12	12	0.200	H
CWR29C-337--H-	4	330	10	100	120	10	12	12	0.180	H
CWR29D-155--A-	6	1.5	1	10	12	6	8	8	4	A
CWR29D-335--A-	6	3.3	1	10	12	6	8	8	6	A
CWR29D-335--B-	6	3.3	1	10	12	6	8	8	3.2	B
CWR29D-475--A-	6	4.7	1	10	12	6	8	8	6	A
CWR29D-475--C-	6	4.7	1	10	12	6	8	8	2.2	C
CWR29D-685--B-	6	6.8	1	10	12	6	8	8	3.2	B
CWR29D-685--D-	6	6.8	1	10	12	6	8	8	1.5	D
CWR29D-106--B-	6	10	1	10	12	6	8	8	3.2	B
CWR29D-106--E-	6	10	1	10	12	8	10	12	1	E
CWR29D-156--B-	6	15	1	10	12	8	10	10	3.2	B

See footnote at the end of table.

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TABLE II. Style CWR29 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR29D-156--D-	6	15	1	10	12	8	10	12	1.7	D
CWR29D-156--E-	6	15	1	10	12	8	10	12	0.9	E
CWR29D-226--D-	6	22	1	10	12	6	8	8	1.7	D
CWR29D-226--E-	6	22	2	20	24	8	10	12	1	E
CWR29D-226--F-	6	22	2	20	24	8	10	12	0.6	F
CWR29D-336--E-	6	33	2	20	24	6	8	8	1	E
CWR29D-476--F-	6	47	3	30	36	8	10	12	1	F
CWR29D-476--G-	6	47	3	30	36	10	12	12	0.275	G
CWR29D-686--F-	6	68	4	40	48	10	12	12	0.400	F
CWR29D-686--G-	6	68	4	40	48	10	12	12	0.25	G
CWR29D-686--H-	6	68	4	40	48	10	12	12	0.180	H
CWR29D-107--G-	6	100	6	60	72	10	12	12	0.275	G
CWR29D-157--G-	6	150	10	100	120	10	12	12	0.275	G
CWR29D-227--H-	6	220	10	100	120	10	12	12	0.180	H
CWR29D-337--H-	6	330	20	200	240	10	12	12	0.18	H
CWR29F-105--A-	10	1	1	10	12	6	8	8	5	A
CWR29F-225--A-	10	2.2	1	10	12	6	8	8	6	A
CWR29F-225--B-	10	2.2	1	10	12	6	8	8	3.2	B
CWR29F-335--A-	10	3.3	1	10	12	6	8	8	6	A
CWR29F-335--C-	10	3.3	1	10	12	6	8	8	2.2	C
CWR29F-475--B-	10	4.7	1	10	12	6	8	8	3.2	B
CWR29F-475--C-	10	4.7	1	10	12	6	8	8	2.2	C
CWR29F-475--D-	10	4.7	1	10	12	6	8	8	1.5	D
CWR29F-685--B-	10	6.8	1	10	12	6	8	8	3.2	B
CWR29F-685--C-	10	6.8	1	10	12	6	8	8	2.2	C
CWR29F-685--D-	10	6.8	1	10	12	6	8	8	1.7	D
CWR29F-685--E-	10	6.8	1	10	12	6	8	8	1	E
CWR29F-106--B-	10	10	1	10	12	8	10	10	3.2	B
CWR29F-106--C-	10	10	1	10	12	6	8	8	2.2	C
CWR29F-106--D-	10	10	1	10	12	6	8	8	1.3	D
CWR29F-106--E-	10	10	1	10	12	6	8	8	1	E
CWR29F-156--D-	10	15	2	20	24	6	8	8	1.7	D
CWR29F-156--E-	10	15	2	20	24	8	10	10	0.9	E
CWR29F-156--F-	10	15	2	20	24	8	8	10	0.7	F
CWR29F-226--E-	10	22	3	30	36	8	10	10	0.6	E
CWR29F-336--F-	10	33	3	30	36	8	10	10	0.4	F
CWR29F-336--G-	10	33	3	30	36	10	12	12	0.275	G
CWR29F-476--F-	10	47	4	40	48	10	12	12	0.4	F
CWR29F-476--G-	10	47	4	40	48	10	12	12	0.25	G
CWR29F-476--H-	10	47	5	50	60	10	12	12	0.180	H
CWR29F-686--G-	10	68	6	60	72	10	12	12	0.275	G
CWR29F-107--G-	10	100	10	100	120	10	12	12	0.275	G
CWR29F-107--H-	10	100	10	100	120	10	12	12	0.180	H
CWR29F-157--H-	10	150	15	150	180	10	12	12	0.180	H
CWR29F-157--X-	10	150	15	150	180	10	12	12	0.065	X
CWR29F-227--H-	10	220	20	200	240	10	12	12	0.180	H
CWR29H-684--A-	15	0.68	1	10	12	6	8	8	6.	A
CWR29H-105--A-	15	1	1	10	12	6	8	8	7.5	A
CWR29H-155--A-	15	1.5	1	10	12	6	8	8	7.5	A
CWR29H-155--B-	15	1.5	1	10	12	6	8	8	3.2	B
CWR29H-225--A-	15	2.2	1	10	12	6	8	8	7.5	A
CWR29H-225--C-	15	2.2	1	10	12	6	8	8	2.2	C
CWR29H-335--B-	15	3.3	1	10	12	6	8	8	3.6	B
CWR29H-335--D-	15	3.3	1	10	12	6	8	8	1.7	D

See footnote at the end of table.

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TABLE II. Style CWR29 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR29H-475--B-	15	4.7	1	10	12	6	8	8	2.	B
CWR29H-475--C-	15	4.7	1	10	12	6	8	8	2.2	C
CWR29H-475--D-	15	4.7	1	10	12	6	8	8	2	D
CWR29H-475--E-	15	4.7	1	10	12	6	8	8	1.2	E
CWR29H-685--D-	15	6.8	1	10	12	6	8	8	2	D
CWR29H-685--E-	15	6.8	1	10	12	8	10	12	0.9	E
CWR29H-106--D-	15	10	2	20	24	6	8	8	2	D
CWR29H-106--E-	15	10	2	20	24	6	8	8	1.2	E
CWR29H-106--F-	15	10	2	20	24	6	8	8	0.667	F
CWR29H-156--E-	15	15	2	20	24	6	8	8	1.2	E
CWR29H-156--F-	15	15	2	20	24	8	10	10	0.8	F
CWR29H-226--F-	15	22	3	30	36	8	10	10	0.8	F
CWR29H-226--G-	15	22	4	40	48	6	8	8	0.275	G
CWR29H-336--F-	15	33	5	50	60	6	8	8	0.8	F
CWR29H-336--G-	15	33	6	60	72	8	10	10	0.275	G
CWR29H-336--H-	15	33	5	50	60	8	8	10	0.18	H
CWR29H-476--G	15	47	10	100	120	8	10	10	0.275	G
CWR29H-476--H-	15	47	10	100	120	8	10	10	0.18	H
CWR29H-686--G-	15	68	10	100	120	8	10	10	0.275	G
CWR29H-686--H-	15	68	10	100	120	8	10	10	0.18	H
CWR29H-107--H-	15	100	15	150	180	10	12	12	0.18	H
CWR29J-474--A-	20	0.47	1	10	12	8	8	10	7.5	A
CWR29J-684--A-	20	0.68	1	10	12	6	8	8	7.5	A
CWR29J-684--B-	20	0.68	1	10	12	6	8	8	5.6	B
CWR29J-105--A-	20	1	1	10	12	6	8	8	7.5	A
CWR29J-105--B-	20	1	1	10	12	6	8	8	4.8	B
CWR29J-155--B-	20	1.5	1	10	12	6	8	8	3.6	B
CWR29J-155--C-	20	1.5	1	10	12	6	8	8	2.4	C
CWR29J-225--B-	20	2.2	1	10	12	6	8	8	3.6	B
CWR29J-225--D-	20	2.2	1	10	12	6	8	8	1.7	D
CWR29J-335--D-	20	3.3	1	10	12	6	8	8	2	D
CWR29J-335--E-	20	3.3	1	10	12	6	8	8	1.2	E
CWR29J-475--E-	20	4.7	1	10	12	6	8	8	1.7	E
CWR29J-685--E-	20	6.8	2	20	24	6	8	8	1.5	E
CWR29J-685--F-	20	6.8	2	20	24	6	8	8	0.7	F
CWR29J-106--E-	20	10	2	20	24	6	8	8	1.5	E
CWR29J-106--F-	20	10	2	20	24	6	8	8	0.8	F
CWR29J-156--F-	20	15	3	30	36	6	8	8	0.8	F
CWR29J-156--G-	20	15	3	30	36	6	8	8	0.275	G
CWR29J-226--G-	20	22	4	40	48	6	8	8	0.625	G
CWR29J-226--H-	20	22	4	40	48	6	8	8	0.18	H
CWR29J-336--H-	20	33	6	60	72	8	10	10	0.18	H
CWR29J-476--H-	20	47	10	100	120	8	10	10	0.18	H
CWR29J-476--X-	20	47	10	100	120	8	10	10	0.11	X
CWR29K-334--A-	25	0.33	1	10	12	6	8	8	7.5	A
CWR29K-474--A-	25	0.47	1	10	12	6	8	8	7.5	A
CWR29K-684--B-	25	0.68	1	10	12	6	8	8	4	B
CWR29K-105--B-	25	1	1	10	12	6	8	8	4	B
CWR29K-105--C-	25	1	1	10	12	6	8	8	2.6	C
CWR29K-155--D-	25	1.5	1	10	12	6	8	8	1.7	D
CWR29K-225--D-	25	2.2	1	10	12	6	8	8	2	D
CWR29K-225--E-	25	2.2	1	10	12	6	8	8	1	E
CWR29K-335--E-	25	3.3	1	10	12	6	8	8	1.2	E
CWR29K-475--F-	25	4.7	2	20	24	6	8	8	0.7	F

See footnote at the end of table.

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TABLE II. Style CWR29 characteristics - Continued.

Pin 1/	DC rated voltage (+85°C) (Volts)	Cap. (nom) (∝F)	DC Leakage (max) (∝A)			Dissipation factor (max) (percent)			Max ESR 100 kHz +25°C (Ohms)	Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C		
CWR29K-685--F-	25	6.8	2	20	24	6	8	8	0.8	F
CWR29K-685--G-	25	6.8	2	20	24	6	8	8	0.3	G
CWR29K-106--G-	25	10	3	30	36	6	8	8	0.35	G
CWR29K-156--G-	25	15	4	40	48	6	8	8	0.35	G
CWR29K-156--H-	25	15	4	40	48	6	8	8	0.2	H
CWR29K-226--G-	25	22	6	60	72	6	8	8	0.35	G
CWR29K-226--H-	25	22	6	60	72	6	8	8	0.18	H
CWR29K-226--X-	25	22	6	60	72	6	8	8	0.16	X
CWR29K-336--H-	25	33	10	100	120	8	10	10	0.18	H
CWR29K-336--X-	25	33	10	100	120	8	10	10	0.13	X
CWR29M-224--A-	35	0.22	1	10	12	6	8	8	12	A
CWR29M-334--A-	35	0.33	1	10	12	6	8	8	12	A
CWR29M-474--B-	35	0.47	1	10	12	6	8	8	6.8	B
CWR29M-684--C-	35	0.68	1	10	12	6	8	8	4	C
CWR29M-105--D-	35	1	1	10	12	6	8	8	2.2	D
CWR29M-155--E-	35	1.5	1	10	12	6	8	8	1.3	E
CWR29M-335--F-	35	3.3	1	10	12	6	8	8	0.7	F
CWR29M-475--G-	35	4.7	2	20	24	6	8	8	0.375	G
CWR29M-685--G-	35	6.8	3	30	36	6	8	8	0.375	G
CWR29M-685--H-	35	6.8	3	30	36	6	8	8	0.5	H
CWR29M-106--H-	35	10	4	40	48	8	10	10	0.5	H
CWR29M-156--X-	35	15	6	60	72	6	8	8	0.19	X
CWR29N-104--A-	50	0.1	1	10	12	6	8	8	12	A
CWR29N-154--A-	50	0.15	1	10	12	6	8	8	12	A
CWR29N-224--B-	50	0.22	1	10	12	6	8	8	6.8	B
CWR29N-334--B-	50	0.33	1	10	12	6	8	8	4.8	B
CWR29N-474--C-	50	0.47	1	10	12	6	8	8	3.2	C
CWR29N-684--D-	50	0.68	1	10	12	6	8	8	2.3	D
CWR29N-105--E-	50	1	1	10	12	6	8	8	1.7	E
CWR29N-155--F-	50	1.5	1	10	12	6	8	8	1.1	F
CWR29N-225--F-	50	2.2	2	20	24	6	8	8	0.7	F
CWR29N-335--G-	50	3.3	2	20	24	6	8	8	0.5	G
CWR29N-475--H-	50	4.7	3	30	36	6	8	8	0.5	H

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, product level designator, and surge current option letter. If surge current is not required, the last dash shall be replaced with the letter Z.

Custodians:
Army - CR
Navy - EC
Air Force - 11

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99
NASA - NA

Preparing activity:
Army - CR

Agent:
DLA - CC

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